

Udayan Atreya

San Jose, CA | 669-340-6033 | udayan.atreya@gmail.com | [LinkedIn](#) - /uatreya | [Github](#) | [Portfolio](#)

EDUCATION

Master of Science, Computer Science

May 2026

San Jose State University, San Jose, US

Coursework: Applied Deep Learning, Retrieval Augmented Generation (RAG), Biometric Security with AI, Multimodal AI Systems.

Bachelor of Technology, Computer Science & Engineering

May 2020

Manipal University, Jaipur, India

Coursework: Data Structures and Algorithms, Relational Database Management Systems, Object-Oriented Programming.

TECHNICAL SKILLS

AI/ML Frameworks: PyTorch, TensorFlow, HuggingFace, LangChain, LangGraph, RAGAS, FastAPI

Multimodal & Generative AI: LLMs (GPT-4o, Gemini, Llama, DeepSeek), Whisper (speech/audio), CLIP, diffusion models

AI Agent Systems: Agentic pipelines, tool use, planning & reasoning loops, multi-agent collaboration, semantic routing

Cloud & Infrastructure: AWS (S3, Lambda, API Gateway, DynamoDB), MongoDB, Pinecone, ChromaDB, CI/CD

Languages: Python, Java, C#, SQL

PROJECT EXPERIENCE

Multimodal Evaluator-Optimizer AI Agent System (*LangChain, GPT-4o, Gemini, Llama-3, Whisper, ChromaDB*) [\[GitHub Link\]](#)

Aug 2025 – Mar 2026

- Architected a multimodal AI agent system (text and audio via Whisper) with an Evaluator-Optimizer reasoning loop, **reducing hallucination by 25%** and **improving accuracy by 18%** over baselines for Language-Based Video Reasoning.
- Engineered multi-agent collaboration with LLM-as-a-judge (Llama-3-70b) across 500+ STEM QA pairs, quantifying Faithfulness, Answer Relevancy, Context Precision, and Semantic Similarity.
- Optimized latency-accuracy trade-off via Dynamic Chunking and MMR search, **reducing context redundancy by 40%** while maintaining **90%+ semantic coverage**; lightweight models achieved **98% parity vs. SOTA** at **50% lower storage cost**.

Agentic AI Observability Platform with Tool Use & Multi-Agent Coordination (*LangGraph, Gemini 1.5, FastAPI, MongoDB, AWS S3*) [\[GitHub Link\]](#)

Jun 2025 – Jul 2025

- Deployed a production-ready AI agent system with tool use, semantic routing, and real-time telemetry (Prometheus, Grafana), achieving **95% query classification accuracy** and **reducing manual triage by 30%**.
- Implemented multi-agent collaboration with human-in-the-loop validation and role-based policy enforcement, **preventing 100% of unauthorized state changes**; built hybrid retrieval engine **cutting root-cause analysis time by 75%** (2m to 30s).

WORK EXPERIENCE

Software Engineer, Accenture, Bengaluru, India

Dec 2021 – Jul 2024

Associate Software Engineer, Accenture, Bengaluru, India

Nov 2020 – Nov 2021

- Built and deployed a production-grade ChatOps AI platform integrating natural language understanding with agentic automation workflows via serverless microservices, **reducing operational overhead by 40%**.
- Developed an analytics-driven recommendation engine with weighted scoring and personalization logic for 100+ automation assets, **reducing asset discovery time by 30%**.
- Architected scalable notification microservices on **AWS (API Gateway, Lambda, DynamoDB)**, achieving **20% increase in DAU** and **reducing retrieval latency by 37%** via redesigned access patterns.
- Delivered a code vulnerability remediation system using serverless architecture (AWS Lambda, S3, EventBridge, Azure DevOps), safeguarding **400+ applications**; designed server patching solution **reducing MTTR by 80%**.